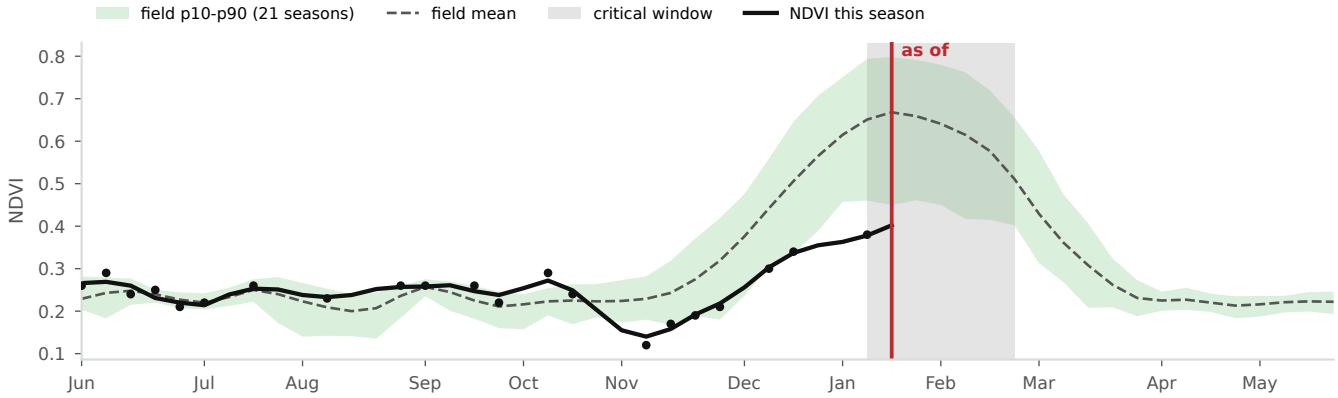
Loan memo: Campo Santa Elena S.A.

L004 · Santa Fe · Centro, Argentina · soy · as of 18 Jan 2023 (week 33)

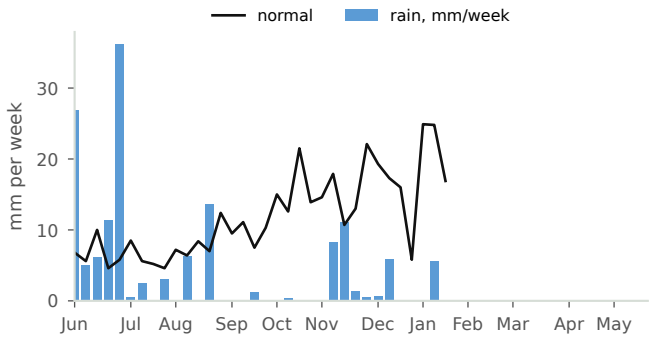
SEVERE score 100/100 · confidence 67% · Stage 2 (critical climate alert (level Severe))

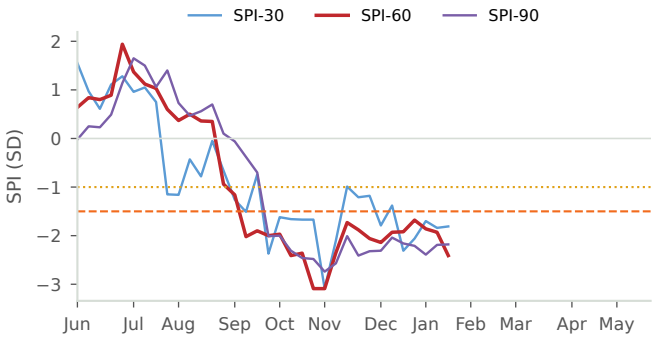
Loan / farmer	**L004 / F004**	Borrower	**Campo Santa Elena S.A.**
Product	**Crop input loan**	Amount and rate	**$105,000 at 8.91% p.a.**
Originated / maturity	**24 Sep 2022 / 24 Apr 2023**	Outstanding (EAD)	**$105,000**
Field	**soy, 330 ha; sown 08 Nov 2022**	Crop stage now	**Flowering (weight 1.00)**
Collateral	**warrant on grain, $120,500**	BCRA / past due	**1 / 0**
Insurance	**none on file**	Other-bank debt	**$49,000**

Greenness (NDVI) against this field's own history

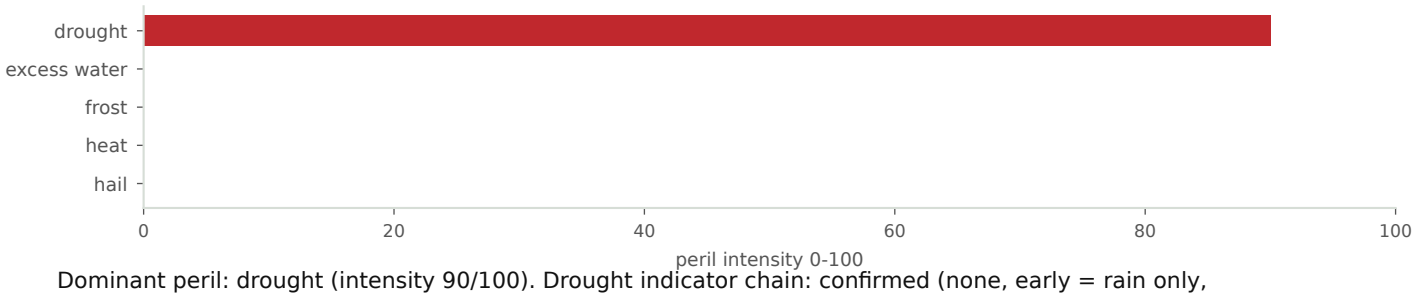


Rainfall and standardised precipitation index





Stress perils, drought cascade and confidence

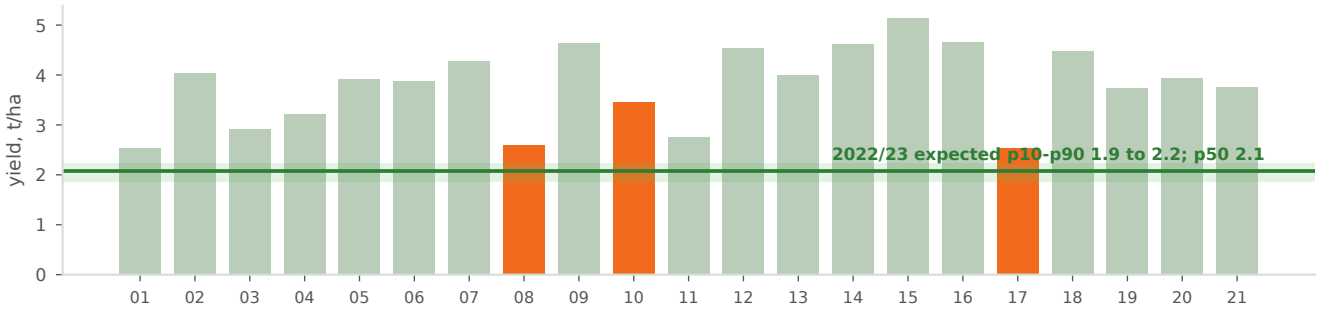


Dominant peril: drought (intensity 90/100). Drought indicator chain: confirmed (none, early = rain only, active = plus soil moisture, confirmed = plus greenness). Confidence 67%: agreement of signals, location quality and share of cloud-free images (0% this week).

Analog seasons and expected yield

Past season	Distance	Yield vs trend	Yield
2010/11	0.91	-8%	3.46 t/ha
2017/18	1.05	-38%	2.53 t/ha
2008/09	1.05	-30%	2.59 t/ha

Expected 2022/23 yield: p50 2.08 t/ha (-35% vs normal 3.18), p10 1.86 (-42%), p90 2.22 (-30%).



Orange bars are the analog seasons; the green band is the p10-p90 range of the regression for this season.

Credit risk: bank baseline against climate-adjusted

Measure	Bank baseline	Climate-adjusted
PD 12-month	10.36% (origination)	24.68% (2.4x)
IFRS 9 stage	performing, Stage 1	Stage 2 (critical climate alert (level Severe))
LGD	37.8%	45.9%
EAD	$105,000	$105,000
ECL base scenario	$4,338	$17,210
ECL adverse scenario (yield p10)	-	$20,387
ECL severe scenario (worst analog)	-	$20,387
ECL probability-weighted	**$4,338**	**$18,481 (+$14,143)**

Explanation

Soy is in flowering (stage weight 1.0). Drought confirmed: 60-day rain is -2.4 SD from normal (SPI-60), soil moisture is at the 5th percentile and NDVI is 2.1 SD below this field's own history. Expected yield -35% vs normal (range -42% to -30%). Closest past season: 2010/11 (yield -8%). Risk level Severe (score 100, confidence 67%).

Recommendation

Confirm Stage 2 classification.

Critical alert (score 100); PD is 2.4x its origination level.

Prepared from synthetic data. Numbers are model output for a credit officer to review; no action is taken automatically.